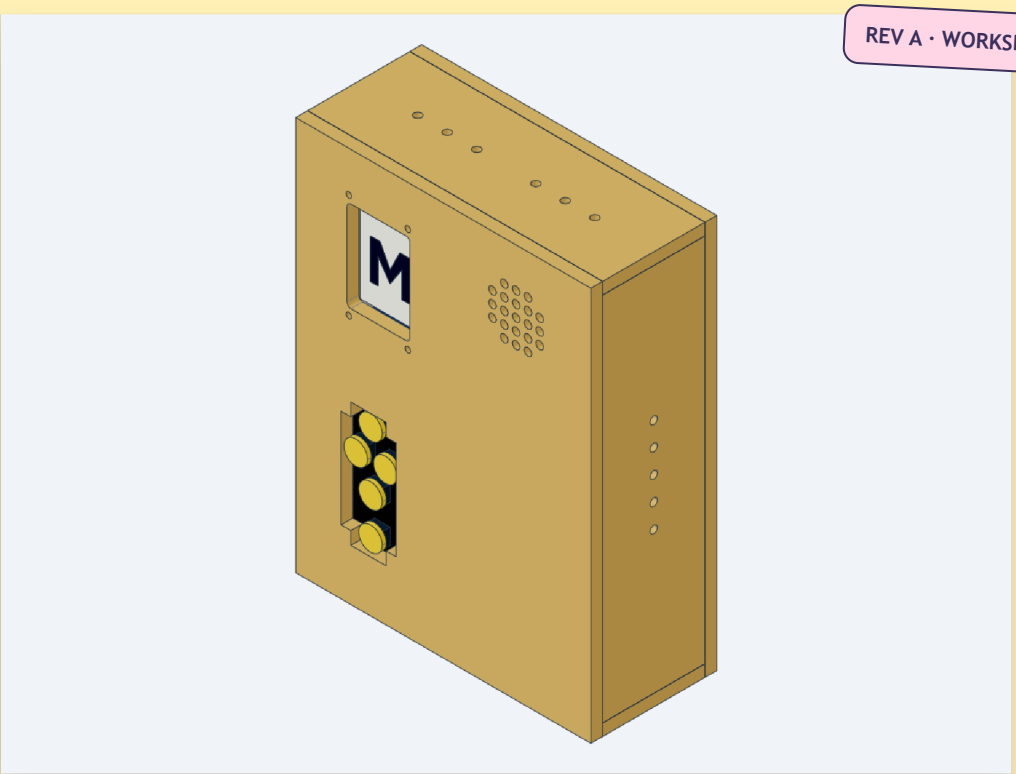
FABRICATION & HARDWARE WORKSHOP GUIDE

# Alfa Radio
# Plywood Box & Raspi 3 Guide

Complete step-by-step instructions for fabricating the 6 mm birch plywood enclosure, configuring the Raspberry Pi 3 hardware, wiring the complete 27-lead atlas, and two sets of 1:1 printable cutting blueprints (Original Box & Little House Radio Case).

REV A · WORKSHOP READY



| OUTER DIMENSIONS | PRIMARY MATERIAL | CORE CONTROLLER | AUDIO & DISPLAY |
|---|---|---|---|
| 156 × 200 × 64 mm | 6 mm Birch Plywood | Raspberry Pi 3 B / B+ | I²S 3W + 2.0" IPS LCD |

**Safety & Printing Notice:** Both blueprint sets (Attachments 1 and 2) are exact 1:1 scale (Actual Size / 100%). Measure the 100 mm calibration line on each sheet before cutting. Never solder or change wiring with power connected.

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## Project Architectural Concept

Alfa is an autonomous offline educational radio designed for toddlers and young children. It features physical tactile control (five sturdy arcade-style buttons), a 2.0-inch color IPS screen, rich audio output, and a self-contained battery power system. The plywood enclosure provides an organic, repairable chassis that protects internal electronics while allowing tool-assisted opening for maintenance.

**Two Enclosure Options Included:**

- **Original Plywood Case (Attachment 1, Pages 9-13):** Compact rectangular box (150 × 200 × 64 mm) using 4 standard A4 sheets and straight butt joints.
- **Little House Radio Case (Attachment 2, Pages 14-18):** Gable house silhouette with dual split-opening roof wings on brass hinges, magnetic closure, and a 22 mm beech dowel carry handle.

Both designs accommodate the identical internal electronics, Raspberry Pi 3 hardware, and 27-wire connection atlas.

# 1. List of Items Needed (Bill of Materials)

Every component required to build and wire the complete Alfa offline radio with the plywood enclosure:

## 1.1. Wood & Structural Materials

| Item | Specification | Qty | Purpose |
|---|---|---|---|
| **Birch Plywood** | 6 mm thickness, Grade BB/BB or B/BB | 4 sheets | Four 210 × 297 mm (A4) blanks or 1× 600×300 mm sheet |
| **Wood Glue** | PVA / Aliphatic resin (e.g. Titebond II / D3) | 1 bottle | Permanent frame butt joints & laminated corner blocks |
| **Sandpaper** | 120, 180, and 240 grit sheets | 1 pack | Deburring edges and opening smoothing |
| **Foam & Velcro** | 3 mm high-density foam + adhesive Velcro | ~15 cm | Battery retention pad and non-slip mounting |

## 1.2. Fasteners, Spacers & Standoffs

| Item | Specification | Qty | Purpose |
|---|---|---|---|
| **Rear Cover Screws** | M3 × 25 mm machine screws (pan/countersunk) | 4 | Secures removable rear cover into threaded corner blocks |
| **Captive Hex Nuts** | M3 standard steel hex nuts | 4 | Embedded into inner face of four laminated corner blocks |
| **Switch Clamp Screws** | M3 × 35 mm machine screws + washers + nuts | 2 | Fastens side switch cover and inside retaining clamp |
| **Switch Spacers** | 12 mm M3 insulating nylon/brass spacers | 2 | Maintains correct spacing between side wall and switch clamp |
| **Pi Standoffs** | 8 mm M2.5 brass/nylon standoffs + M2.5 screws | 4 | Mounts Raspberry Pi 3 portrait to inside of rear cover |
| **LCD Spacers** | 4 mm M2.5 or M3 insulating spacers | 4 | Sets screen flush against 6 mm front panel window |
| **Module Hardware** | Small M2/M2.5 screws, nuts & insulating washers | 8 | Mounting MAX98357A, ADS1115, keypad & divider board |

## 1.3. Electronics & Development Modules

| Item | Specification | Qty | Purpose |
|---|---|---|---|
| **Raspberry Pi 3** | Raspberry Pi 3 Model B (or 3 B+) + 16GB MicroSD | 1 | Main system computer running Alfa Linux player |
| **Waveshare LCD** | 2.0-inch IPS LCD, 240×320, SPI ST7789, SKU 17344 | 1 | Character drawing and visual learning display |
| **Audio Amplifier** | MAX98357A I²S 3.2W Class D mono breakout | 1 | Drives speaker directly from Raspberry Pi digital I²S audio |
| **Analog ADC** | ADS1115 16-bit 4-channel I²C ADC breakout | 1 | Reads analog stepped voltage from 5-button keypad |
| **5-Button Keypad** | AD-Key module with 5 tactile switches, 1 analog OUT | 1 | Tactile child controls (Left, Right, Up, Down, Center) |
| **Speaker** | 3 W / 4 Ω passive dynamic speaker (40 × 40 mm) | 1 | Clear vocal letter pronunciation and sound cues |
| **Resistor Divider** | 2× 10 kΩ 1/4W resistors + 16×20 mm perfboard | 1 | Halves keypad 0–5V signal down to safe 0–2.5V for ADC |
| **Power Bank** | Baseus 5000 mAh ultra-slim (69 × 106 × 14 mm) | 1 | Internal rechargeable 5V power supply |
| **Switched USB Lead** | Micro-USB cable with latching push-switch | 1 | Physical power disconnect switch on left side |

# 2. How to Assemble the Plywood Box

Follow this verified workshop order to build the enclosure accurately and ensure all internal components fit perfectly.

## 2.1. Cut List Summary (Four A4 Sheets of 6 mm Plywood)

| Sheet | Part Name | Dimensions | Qty | Features |
|---|---|---|---|---|
| Sheet 1 | Front Panel | 150 × 200 × 6 mm | 1 | 32×46 LCD cutout, 28×63 keypad cutout, 21-hole speaker grille |
| Sheet 2 | Rear Cover | 150 × 200 × 6 mm | 1 | 4× M3 screw holes, 10 cooling vents, 49×58 Pi mount pattern |
| Sheet 3 | Left & Right Walls | 52 × 188 × 6 mm | 2 | 5× vent holes per wall; left wall has 26×44 mm switch window |
| Sheet 3 | Top & Bottom | 150 × 52 × 6 mm | 2 | 6× vent holes per panel; spans full 150 mm box width |
| Sheet 4 | Switch Cover | 40 × 72 × 6 mm | 1 | Outer cover with Ø16 mm hole for push-button actuator |
| Sheet 4 | Inside Switch Clamp | 22 × 64 × 6 mm | 1 | Inner clamping bar securing switch body against left wall |
| Sheet 4 | Corner Block Pads | 12 × 12 × 6 mm | 8 | Glued in pairs to make four 12 mm corner blocks with M3 nuts |

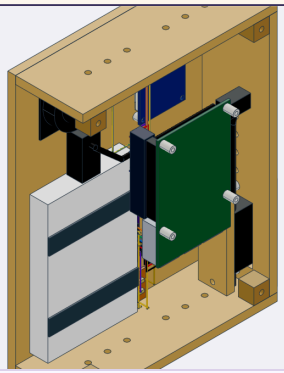
## 2.2. Step-by-Step Woodworking Instructions

1. **Print and Verify Templates:** Print Sheets 1–4 from Attachment 1 at **100% (Actual Size)**. Measure the 100 mm calibration line with a steel ruler. If it is not exactly 100 mm, adjust printer scale settings. Fix the templates to the plywood with removable spray adhesive or tape.
2. **Cut Out Outer Panels:** Cut the outer borders using a fine-toothed hand saw or jigsaw, cutting slightly to the waste side of the line, then sanding flat to the template boundary.
3. **Cut Internal Apertures:** For closed cutouts (LCD window 32 × 46 mm, stepped keypad opening 28 × 63 mm, side switch slot 26 × 44 mm), drill a 4 mm starter hole inside the waste area, feed your coping or fret saw blade through, and carefully cut out the opening. Pay close attention to the stepped keypad tabs—do not cut away the four corner support tabs.
4. **Drill Ventilation & Mounting Holes:**
   - Speaker grille: drill 21 holes (Ø4.0 mm) on a 6 mm grid at X112 Y32.
   - Cooling vents: drill ten Ø4.0 mm holes on rear cover, five on each side wall, and six on top and bottom panels.
   - Mounting holes: drill four Ø3.0 mm holes for LCD, four Ø3.0 mm holes on rear cover for Pi 3 (49 × 58 mm pattern), and four Ø3.5 mm holes in the rear cover corners for M3 screws.
5. **Fabricate Laminated Corner Blocks:** Glue pairs of 12 × 12 × 6 mm pads together to form four 12 mm thick blocks. Drill a Ø3.5 mm center hole through each block. Press an M3 hex nut into the inward face and secure with a tiny drop of cyanoacrylate (superglue), keeping glue away from the internal threads.
6. **Glue the Main Frame:** Apply wood glue to the ends of the left and right walls (52 × 188 mm) and join them between the top and bottom panels (150 × 52 mm). Use a try square to ensure all corners are exactly 90°. Clamp firmly and allow to cure for 60 minutes.
7. **Attach the Front Panel:** Apply glue along the front perimeter of the rectangular frame and position the front panel (150 × 200 mm) flush with all outer edges. Clamp and let cure.
8. **Install Corner Blocks:** Glue the four threaded corner blocks into the inner rear corners of the box, flush with the rear opening plane. *Do not glue the rear cover!* The rear cover must remain removable.

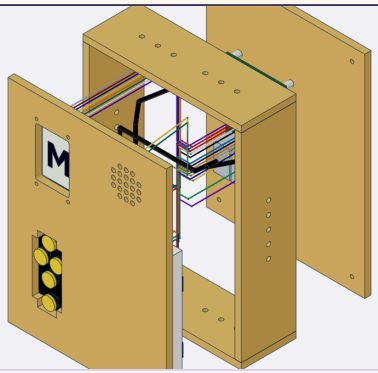
**Provisional Hole Pitch Warning:** Amber markings on templates are placement guides based on standard module drawings. Always place your actual purchased components (LCD, keypad, switch) over the paper templates to verify mounting hole positions before drilling!

# 2.3. Internal Component Installation & Power Setup

The clear inside volume is 138 × 188 × 52 mm. The components are organized in layered depth zones to prevent mechanical interference while maintaining accessibility.



Internal layout: LCD & buttons on front; Pi 3 on rear cover.



Exploded view: Removable rear cover and side switch clamp.

## Step-by-Step Installation Order

9. **Mount Front Panel Electronics:**
   - **LCD:** Place Waveshare 2.0" display into the window from inside using 4 mm nylon spacers. Screen faces the opening. Fasten gently with M2.5/M3 screws.
   - **Speaker:** Center 40 × 40 mm speaker over the 21-hole grille. Secure with corner brackets or foam gasket and small screws.
   - **Keypad:** Seat the 5-button keypad board behind the stepped window. Check that all 5 button caps travel freely through the 6 mm plywood without sticking. Fasten to the corner plywood tabs.
   - **Small Boards:** Mount MAX98357A amplifier and ADS1115 ADC on 4 mm and 3 mm insulating standoffs. Mount the small 16 × 20 mm resistor divider perfboard beside the keypad.
10. **Install Power Bank:** Apply 3 mm foam pad and adhesive Velcro at X73 Y82 (right side of inside front). Place Baseus 5000 mAh battery with its USB-C port facing upward. Fasten two retaining straps around the bank. Ensure the power button on the bank is accessible when the rear cover is opened.
11. **Assemble Side USB Power Switch:** Place the body of the switched micro-USB cable into the 26 × 44 mm side slot in the left wall. Place the inside plywood clamp (22 × 64 mm) over the switch body, separated by two 12 mm M3 spacers. Insert two M3 × 35 mm screws through the outer switch cover (40 × 72 mm), through the wall, through the spacers, and through the clamp. Tighten nuts until the switch is firmly retained without crushing its plastic housing.
12. **Mount Raspberry Pi 3 on Rear Cover:** Thread four 8 mm M2.5 brass standoffs into the rear cover holes. Fasten Raspberry Pi 3 portrait orientation, components facing inward into the box, USB/Ethernet ports facing down/left.
13. **Power Chain Connection:** Connect: Baseus USB-C port → short USB-C to micro-USB cable → switched cable female socket → cable switch body → micro-USB male connector → Raspberry Pi 3 power input.
14. **Signal Wire Service Loop (CRITICAL):** Wire all connections following the wiring atlas. Add **100 mm of extra wire length** on all leads running from the front panel to the Raspberry Pi GPIO header. This allows lifting the rear cover by ~20 mm to disconnect power before full opening.
15. **Initial Testing & Closure:** Test all 5 buttons, display rendering, and audio before closing. Lift and lower the rear cover to verify no wires pinch against the frame or battery. Fasten with four M3 × 25 mm screws into the corner blocks.

**Battery Charging Procedure:** The Baseus power bank shares its single USB-C port for both output and recharging. To recharge: shut down the Pi via software, turn off the side switch, remove the 4 rear screws, open the rear cover 20 mm, unplug the power lead from the bank, and plug in a USB-C charger.

# 3. Raspberry Pi 3 Hardware Setup & Linux Configuration

The Alfa player runs on Raspberry Pi OS (Debian Bookworm or newer, 64-bit Lite recommended). Follow this configuration guide to set up kernel drivers, Device Tree overlays, and audio subsystems.

## 3.1. Kernel Overlays & Configuration ( `/boot/firmware/config.txt` )

Open `/boot/firmware/config.txt` (or `/boot/config.txt` on older OS releases) and add the following configuration lines:

```
# --- Alfa Radio Hardware Configuration ---
# 1. Enable SPI for Waveshare 2.0-inch ST7789 LCD
dtparam=spi=on

# 2. Enable I2C for ADS1115 ADC Keypad Reader
dtparam=i2c_arm=on
dtparam=i2c_arm_baudrate=400000

# 3. Enable MAX98357A I2S DAC Mono Amplifier
dtoverlay=max98357a

# 4. Disable onboard analog headphone audio (prevents ALSA card index confusion)
dtparam=audio=off
```

## 3.2. CRITICAL GPIO PIN CONFLICT RESOLUTION

**PIN CONFLICT WARNING:** Waveshare's standard LCD documentation specifies connecting the display Backlight ( `BL` ) to physical **Pin 12 (GPIO18)**.
However, the MAX98357A I²S digital audio amplifier requires physical **Pin 12 (GPIO18)** as its **Bit Clock (BCLK)**!
If both modules were wired to Pin 12, the audio clock would rapidly toggle the display backlight, causing audio failure and display flickering.

**The Alfa Solution:**

- Physical **Pin 12 (GPIO18)** is dedicated exclusively to the MAX98357A audio Bit Clock ( `BCLK` ).
- Waveshare LCD Backlight ( `BL` ) is moved to physical **Pin 18 (GPIO24)**.
- The Alfa software application ( `app/display.py` ) initializes GPIO24 as the backlight PWM/digital control pin.

## 3.3. Audio Configuration (ALSA & Volume Ceiling)

The MAX98357A is a digital I²S amplifier without hardware volume registers. Software volume scaling is handled by ALSA via `/etc/asound.conf` :

```
pcm.!default {
  type plug
  slave.pcm "softvol"
}
pcm.softvol {
  type softvol
  slave.pcm "hw:0,0"
  control { name "Master"; card 0; }
  max_dB 0.0; min_dB -50.0
}
```

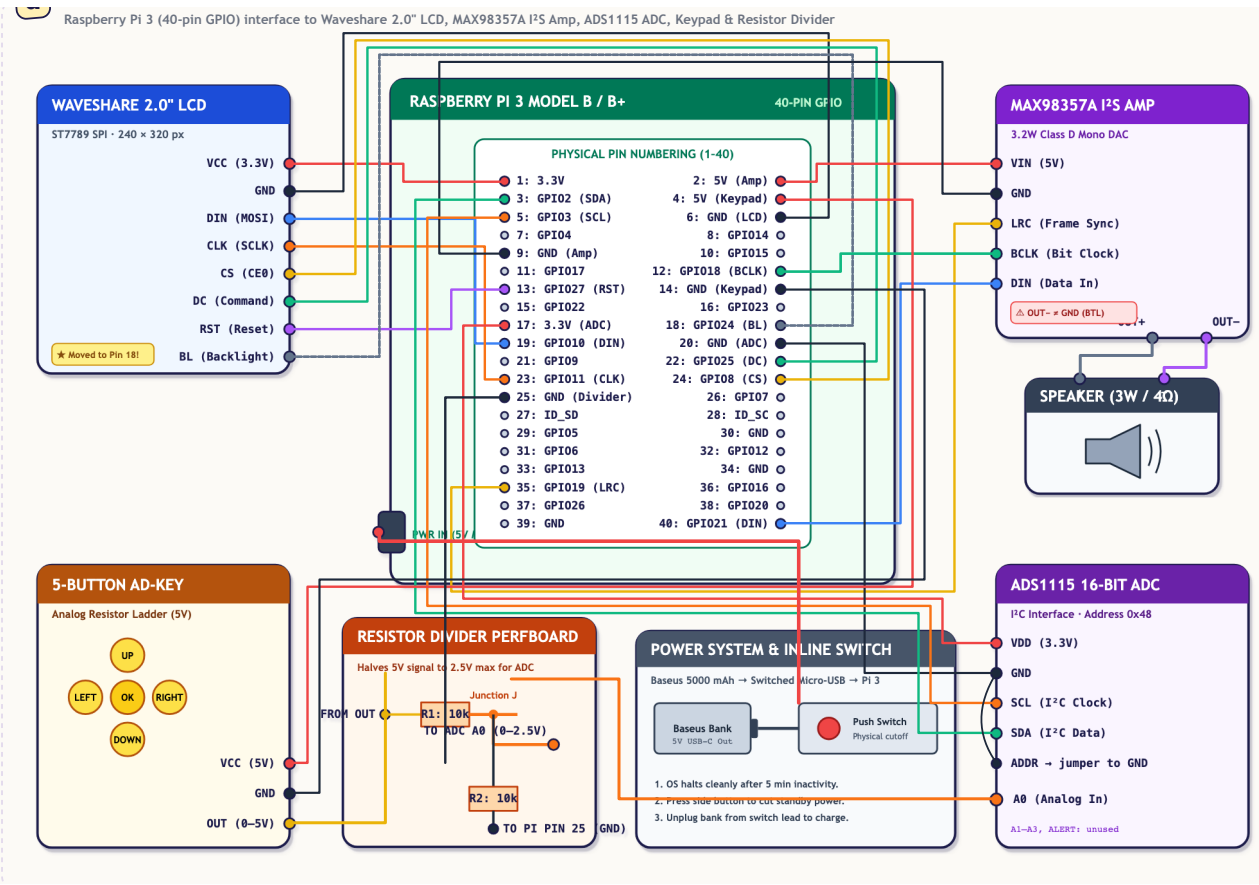
Because the MAX98357A can deliver up to 3.2 Watts into a 4 Ω speaker, set initial volume to ~65% via `alsamixer` to protect children's hearing and avoid speaker clipping.

## 3.4. Inactivity Timer & Power Management

Alfa implements an automatic power-saving routine: after **5 minutes of inactivity**, the player smoothly fades out audio, shows a goodbye screen, and executes a clean Linux shutdown ( `sudo poweroff` ). When the green Pi activity LED stops flashing, the user presses the side inline USB push-switch to disconnect battery power completely.

# 4. Detailed Connection Diagram & System Architecture

This diagram illustrates how all peripheral modules interface with the Raspberry Pi 3 40-pin GPIO header and power distribution.




Complete schematic wiring diagram: Raspberry Pi 3 40-pin GPIO, Waveshare 2" LCD (SPI), MAX98357A (I²S), 3W Speaker, ADS1115 ADC (I²C), 5-Button Keypad, 10k Divider & Battery Power.

## Electrical Principles & Grounding Rules

- **Star Grounding:** Connect module grounds directly to dedicated Pi GND pins (Pin 6, 9, 14, 20, 25). Do not daisy-chain ground wires in a loop to avoid digital switching noise from bleeding into the audio amplifier.
- **Keypad Voltage Divider:** The 5-button AD-Key module operates at 5V, outputting analog signals from 0V to 5V. The Raspberry Pi and ADS1115 ADC logic cannot tolerate 5V on their signal pins. Two 10 kΩ resistors in series halve the voltage to 0–2.5V, well within the ADS1115 3.3V reference window.
- **Speaker Differential Output:** The MAX98357A uses Bridge-Tied Load (BTL) output. Neither `OUT+` nor `OUT−` connects to ground. Connecting `OUT−` to GND will cause an immediate short circuit and destroy the amplifier IC.

**CRITICAL WIRING RULES:**

1. **NEVER connect Speaker (-) to GND.** It connects ONLY to MAX98357A `OUT−`.
2. **NEVER connect Keypad OUT directly to the ADC or Pi without the 10k/10k divider.**
3. **Always double-check pin numbers against physical header positions** before turning on battery power.

# 4.1. Complete 27-Wire Connection Atlas

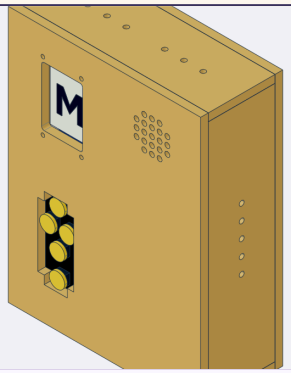
Every wire in the Alfa player with its wire ID, circuit group, endpoints, physical pin numbers, and suggested color:

| Wire | Circuit | From (Component Pin) | To (Destination / Pi Pin) | Signal Function | Color |
|---|---|---|---|---|---|
| W01 | LCD | Waveshare LCD `VCC` | Pin 1 (3.3V) | 3.3V Logic Power | Red |
| W02 | LCD | Waveshare LCD `GND` | Pin 6 (GND) | Ground | Black |
| W03 | LCD | Waveshare LCD `DIN` | Pin 19 (GPIO10) | SPI0 MOSI Data | Blue |
| W04 | LCD | Waveshare LCD `CLK` | Pin 23 (GPIO11) | SPI0 SCLK Clock | Orange |
| W05 | LCD | Waveshare LCD `CS` | Pin 24 (GPIO8) | SPI0 CE0 Chip Select | Yellow |
| W06 | LCD | Waveshare LCD `DC` | Pin 22 (GPIO25) | Data/Command Select | Green |
| W07 | LCD | Waveshare LCD `RST` | Pin 13 (GPIO27) | Display Hardware Reset | Purple |
| W08 | LCD | Waveshare LCD `BL` | Pin 18 (GPIO24) | Backlight PWM (Relocated) | White |
| W09 | Audio | MAX98357A `VIN` | Pin 2 (5V) | 5V Amplifier Power | Red |
| W10 | Audio | MAX98357A `GND` | Pin 9 (GND) | Audio Ground | Black |
| W11 | Audio | MAX98357A `LRC` | Pin 35 (GPIO19) | I²S Frame Clock (PCM_FS) | Yellow |
| W12 | Audio | MAX98357A `BCLK` | Pin 12 (GPIO18) | I²S Bit Clock (PCM_CLK) | Green |
| W13 | Audio | MAX98357A `DIN` | Pin 40 (GPIO21) | I²S Data (PCM_DOUT) | Blue |
| W14 | Speaker | MAX98357A `OUT+` | Speaker (+) Terminal | Positive Audio Phase | Gray |
| W15 | Speaker | MAX98357A `OUT−` | Speaker (-) Terminal | Negative Audio Phase (NOT GND!) | Purple |
| W16 | ADC | ADS1115 `VDD` | Pin 17 (3.3V) | 3.3V ADC Power | Red |
| W17 | ADC | ADS1115 `GND` | Pin 20 (GND) | ADC Ground (ADDR to GND) | Black |
| W18 | ADC | ADS1115 `SCL` | Pin 5 (GPIO3) | I²C Clock (I2C1_SCL) | Orange |
| W19 | ADC | ADS1115 `SDA` | Pin 3 (GPIO2) | I²C Data (I2C1_SDA) | Green |
| W20 | Keypad | AD-Keypad `VCC` | Pin 4 (5V) | 5V Button Ladder Bias | Red |
| W21 | Keypad | AD-Keypad `GND` | Pin 14 (GND) | Button Ladder Ground | Black |
| W22 | Divider | AD-Keypad `OUT` | Divider Perfboard R1 In | Raw 0-5V Keypad Signal | Yellow |
| W23 | Divider | Divider Midpoint (R1/R2) | ADS1115 `A0` Input | Divided 0-2.5V Safe Signal | Orange |
| W24 | Divider | Divider R2 Ground End | Pin 25 (GND) | Divider Ground Return | Black |
| W25 | Power | Switched Micro-USB (+) | Pi Micro-USB Power In (+) | Switched 5V Power Supply | Red |
| W26 | Power | Switched Micro-USB (-) | Pi Micro-USB Power In (-) | Main Power Ground | Black |
| W27 | Power | Baseus USB-C Output | Switched Lead Female Socket | Upstream 5V Battery Connection | Black |

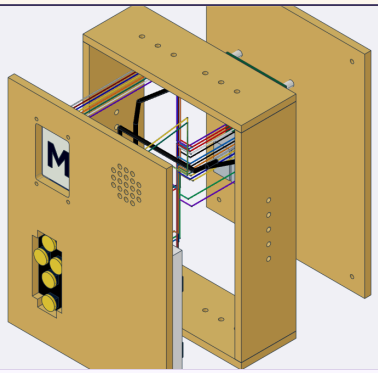
**Wire Preparation Tip:** Twist wire pairs W11/W12/W13 (I²S clocks/data) together, and twist W14/W15 (Speaker outputs) together. Twisting reduces electromagnetic radiation and hum. Ensure all leads connecting to the Raspberry Pi have at least 100 mm service slack.

# Attachment 1: Original Plywood Case Blueprints

This attachment provides the complete set of four 1:1 cutting and drilling stencils for the original rectangular Alfa plywood enclosure (150 × 200 × 64 mm). Fabricated from four A4 sheets of 6 mm birch plywood.



Original case: clean rectangular chassis with side switch cover.

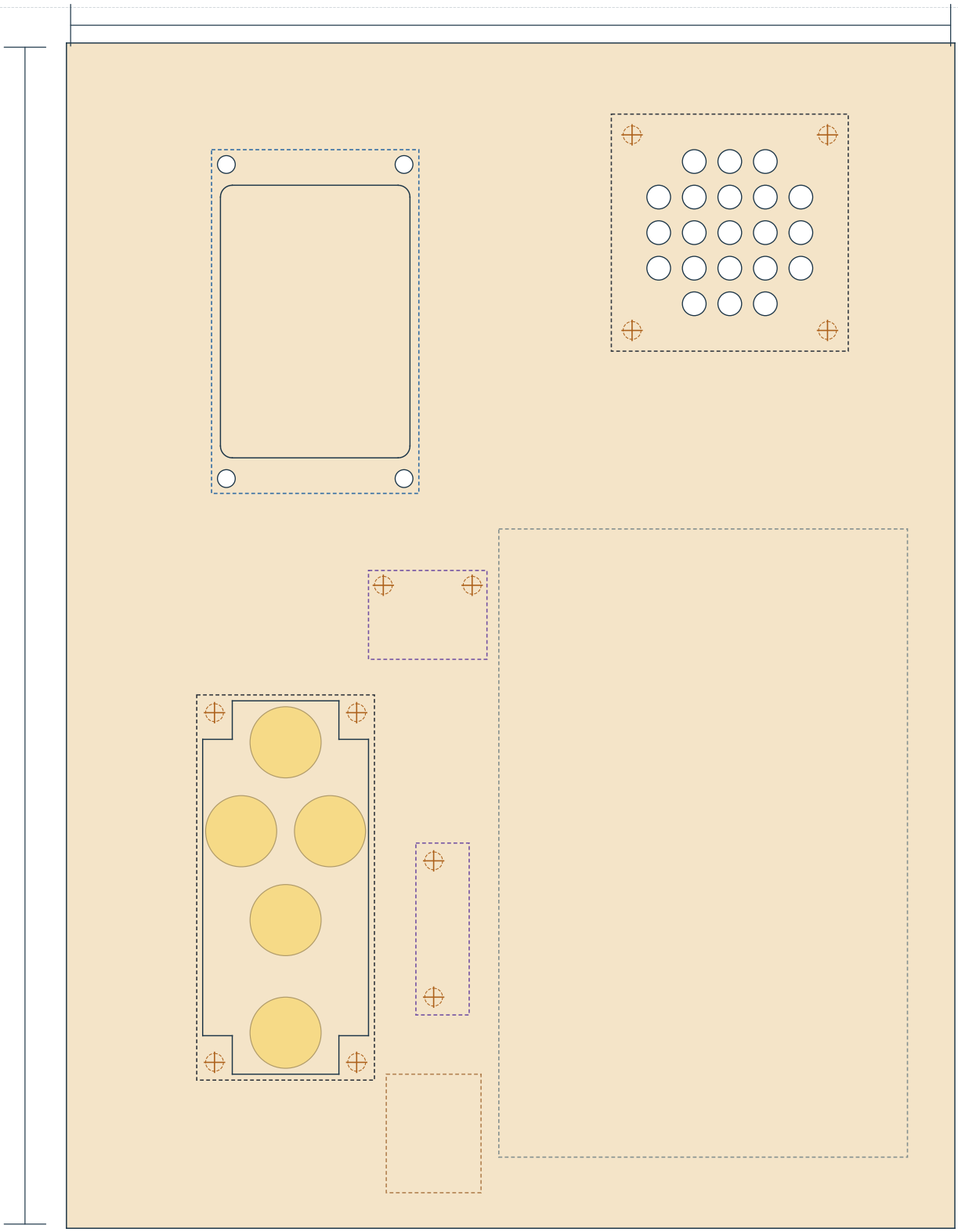


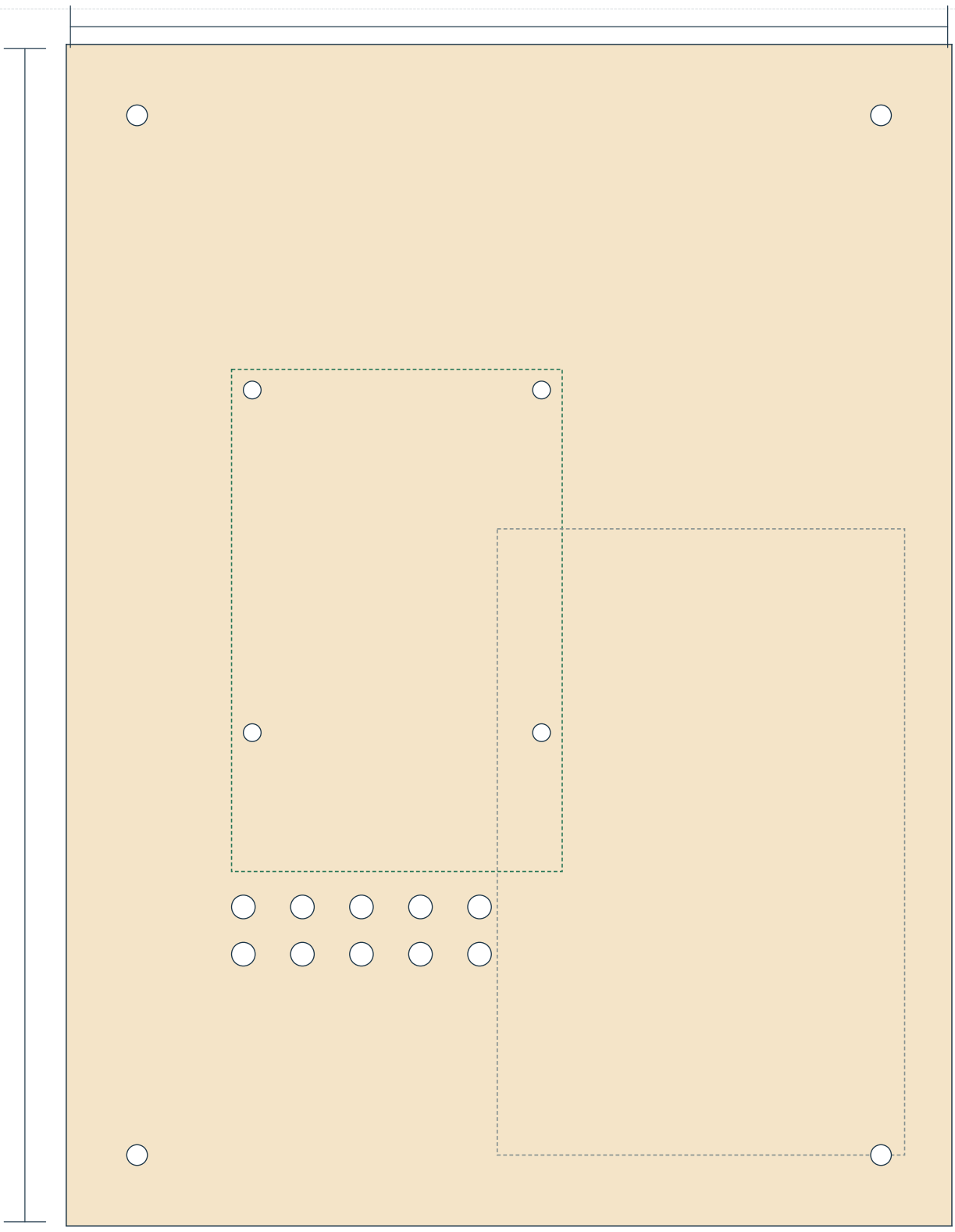
Exploded view: Butt joints, 4 corner blocks, removable rear cover.

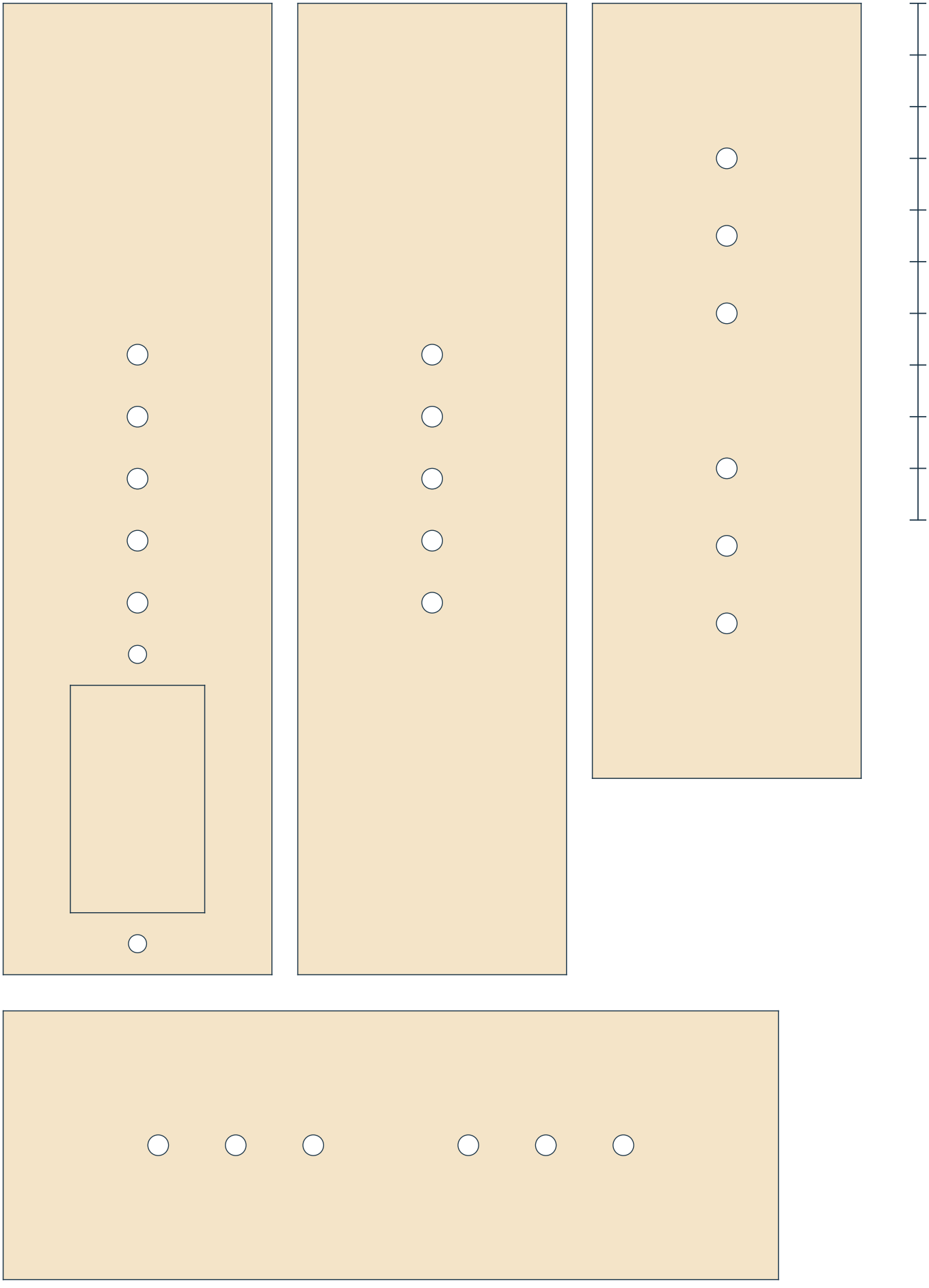
## Blueprint Index (Sheets 1-4)

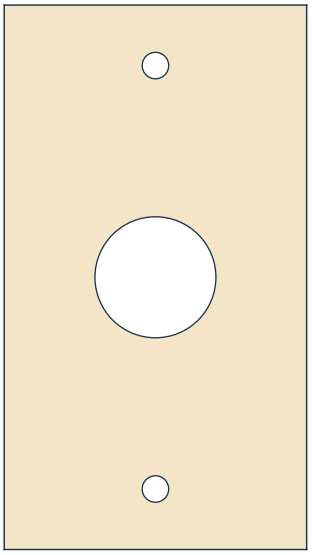
- **Sheet 1 (Page 10):** Front Panel (150 × 200 mm) with LCD window (32 × 46 mm), stepped 5-button keypad cutout (28 × 63 mm), 21-hole speaker grille (Ø4 mm), and 100 mm calibration line.
- **Sheet 2 (Page 11):** Removable Rear Cover (150 × 200 mm) with four corner M3 screw clearance holes, ten cooling vents, Raspberry Pi 3 mounting holes (49 × 58 mm pattern), and 100 mm calibration line.
- **Sheet 3 (Page 12):** Left & Right Walls (52 × 188 mm) with side vents and switch slot; Top & Bottom Panels (150 × 52 mm) with vents; vertical 100 mm calibration gauge.
- **Sheet 4 (Page 13):** Side Switch Cover (40 × 72 mm), Inside Switch Clamp (22 × 64 mm), eight Corner Block Pads (12 × 12 mm), and 100 mm calibration gauge.

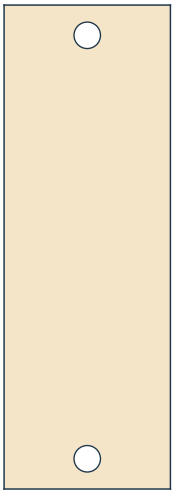
**Printing Scale Calibration:** Every sheet includes a **100 mm calibration line**. You MUST measure this gauge with a ruler before cutting any wood. Set your printer software to **"Actual Size" / 100% scaling**. Never select "Fit to Printable Area" or "Shrink Oversized Pages".

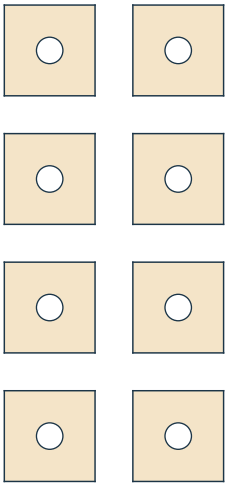








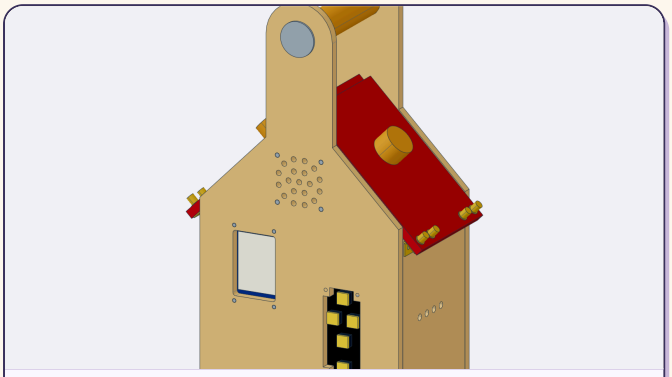




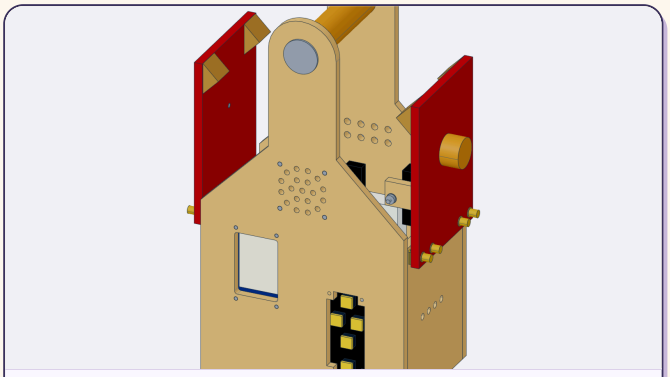


# Attachment 2: Little House Radio Case Blueprints

This attachment provides the complete set of four 1:1 cutting and drilling stencils for the Little House Radio enclosure (178.3 × 278 × 100 mm). Designed around a gable silhouette with a split opening roof on brass hinges and a sturdy 22 mm beech dowel carry handle.



Gable house silhouette with chimney tabs and 22 mm dowel handle.



Split roof opens like wings for battery charging; clicks shut with magnets.

## Architectural Features & Cut List

- **Split Roof Mechanism:** Two identical roof boards (120 × 86 mm) open on 25 × 20 mm brass butt hinges. Overhang lands on the wall at 90° as a positive stop. Neodymium magnets (Ø10 × 3 mm) in wooden wedges click the roof shut.
- **Central Battery Stand:** Baseus 5000 mAh power bank stands upright on the floor against the back gable, held by a wooden fence, with its USB-C port right below the ridge for easy charging access.
- **Internal Layout:** Raspberry Pi 3 mounts to the inside of the left wall; QUARKZMAN push switch mounts high in the back gable.

| Sheet | Part Name | Dimensions | Qty | Features |
|---|---|---|---|---|
| Sheet 1 (P.15) | Front Gable (Part A) | 150 × 278 × 6 mm | 1 | LCD window, button opening, attic speaker grille |
| Sheet 2 (P.16) | Back Gable (Part B) | 150 × 278 × 6 mm | 1 | Removable service panel, switch hole, dowel mount |
| Sheet 3 (P.17) | Floor & Walls (Parts C, D, E) | 88 × 138 mm / 88 × 121.5 mm | 3 | Floor, left/right walls, switch clamp, 8 corner blocks |
| Sheet 4 (P.18) | Roof Boards & Wedges (F, H, J) | 86 × 120.1 mm / 16 × 16 mm | 15 | 2 roof boards, 12 magnet wedges, battery fence |

**Printing Scale Calibration:** Print Sheets 1–4 at **100% (Actual Size)**. Measure the 100 mm calibration gauge with a ruler on every page before making saw cuts.

